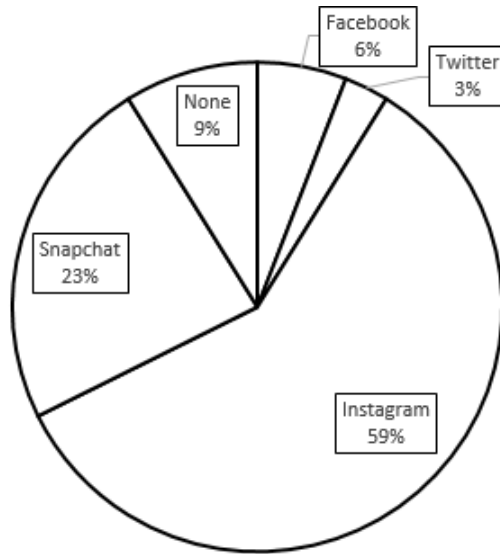


Algebra 1
Unit 12
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

A circle graph is shown, with the results of a survey of 9th grade students about which social media platforms they use, with a margin of error of 1.2%. There are 475 9th graders in a particular school.



1. Based on this survey, what is the estimated number of 9th graders who have an Instagram account?

$$0.59 \cdot 475 \approx 280 \text{ ninth graders}$$

2. Based on this survey, what is the minimum number of 9th graders that would be expected to have a Facebook account?

$$6\% - 1.2\% = 4.8\% \quad 0.048 \cdot 475 \approx 23 \text{ ninth graders}$$

3. Based on this survey, what is the maximum number of 9th graders that would be expected to have a Twitter account?

$$3\% + 1.2\% = 4.2\% \quad 0.042 \cdot 475 \approx 20 \text{ ninth graders}$$

4. Based on this survey, what is the range of 9th graders that would be expected to have a Snapchat account?

$$\begin{aligned} 23\% + 1.2\% &= 24.2\% & 0.242 \cdot 475 &\approx 115 \\ 23\% - 1.2\% &= 21.8\% & 0.218 \cdot 475 &\approx 104 \end{aligned} \quad \text{Approximately 104 to 115.}$$

5. Based on this survey, what is the range of 9th graders that would be expected to have no social media accounts?

$$\begin{aligned} 9\% + 1.2\% &= 10.2\% & 0.102 \cdot 475 &\approx 48 \\ 9\% - 1.2\% &= 7.8\% & 0.078 \cdot 475 &\approx 37 \end{aligned} \quad \text{Approximately 37 to 48}$$

A survey of parents indicated that 62% of parents felt their students spent too much time on social media, with a margin of error of 2%.

6. Out of a group of 2400 parents, how many parents would likely feel that their children spend too much time on social media?

$$0.62 \cdot 2400 = 1,488 \text{ parents} \quad 0.60 \cdot 2400 = 1440 \quad 0.64 \cdot 2400 = 1536$$

1440 to 1536 parents

7. Out of a group of 2400 parents, how many parents would likely **not** feel that their children spend too much time on social media?

$$0.38 \cdot 2400 = 912 \text{ parents} \quad 0.36 \cdot 2400 = 864 \quad 0.40 \cdot 2400 = 960$$

864 to 960 parents

In a survey of 200 families, 32% of families reported that they eat out less than twice a week, 45% report they eat out twice a week, and 23% report they eat out more than twice a week. The survey has a margin of error of 0.8%.

8. What is the expected number of families that eat out twice a week?

$$0.45 \cdot 200 = 90 \text{ families}$$

9. What is the range of families that eat out less than twice a week?

$$\begin{array}{lll} 32 + 0.8 = 32.8\% & 0.328 \cdot 200 \approx 66 & \text{Approximately 62 to 66 families} \\ 32 - 0.8 = 31.2\% & 0.312 \cdot 200 \approx 62 & \end{array}$$

10. What is the range of families that eat out more than twice a week?

$$\begin{array}{lll} 23 + 0.8 = 23.8\% & 0.238 \cdot 200 \approx 47 & \text{Approximately 44 to 47 families} \\ 23 - 0.8 = 22.2\% & 0.222 \cdot 200 \approx 44 & \end{array}$$